

Gas with variable heat capacity — Solution

W26 (2026) — English translation

A1

0.50 pts

Determine the dependence of the internal energy of such a gas on temperature. Consider that at $T = 0$ the internal energy is zero.

Let us write down a small increment of the internal energy of the gas:

$$dU = C_V dT = (C_0 + aT) dT$$

Integrating, we obtain:

Answer: \textbf{Answer:}

$$U = C_0 T + \frac{aT^2}{2}$$

A2

0.30 pts

What is the heat capacity of gas C_P at constant pressure at a given temperature T ?

According to Mayer's equation:

$$C_P = C_V + R \cdot 1 \text{ mol}$$

here and further мы опускаем множитель 1 *mol* in all expressions. From here we get:

Answer: \textbf{Answer:}

$$C_P = C_0 + aT + R$$

A3

0.40 pts

Let an isochoric process occur with a gas, in which the pressure increases from p_1 to p_2 , the volume is equal to V . Find the amount of heat supplied Q_V .

Let's write down I the beginning of thermodynamics:

$$dQ = dU + dA$$

The process is isochoric, so the gas does not do any work, and

$$Q_V = \Delta U = C_0 \Delta T + \frac{a(T_2^2 - T_1^2)}{2}$$

The equation of state of an ideal gas:

$$pV = RT,$$

from where we can find the temperature values at the beginning and at the end of the process:

$$T_1 = \frac{p_1 V}{R}$$

$$T_2 = \frac{p_2 V}{R}$$

We express the answer through the data in the quantity condition:

Answer: \textbf{Answer:}

$$Q_V = \frac{C_0(p_2 - p_1)V}{R} + \frac{a(p_2^2 - p_1^2)V^2}{2R^2}$$

A4

0.40 pts

Let an isobaric process be carried out with a gas, the initial volume is V_1 , the final volume is V_2 , and the pressure is p . Find the amount of heat supplied Q_P .

Let's express the small increment of heat:

$$dQ = C_P dT = (C_0 + aT + R)dT$$

By integrating, we get an answer similar to the previous point:

$$Q_P = (C_0 + R)\Delta T + \frac{a(T_2^2 - T_1^2)}{2}$$

Through the equation of state we get the temperatures at the beginning and end of the process:

$$T_1 = \frac{pV_1}{R}$$

$$T_2 = \frac{pV_2}{R}$$

Let's express the answer through the data in the quantity condition:

Answer: \textbf{Answer:}

$$Q_P = \frac{(C_0 + R)(V_2 - V_1)p}{R} + \frac{a(V_2^2 - V_1^2)p^2}{2R^2}$$

A5

1.50 pts

Consider a cycle 123, consisting of a process 1 – 2, in which the pressure is directly proportional to the volume, an isochore 2 – 3 and an isobar 3 – 1. Moreover, it is known that $V_2 = 2V_1$, and the heat capacities at constant volumes at points 1 and 2 are related by the relation $C_{V2} = 1.5C_{V1}$. Determine the efficiency of this cycle. Find the numerical value. Meaning $C_0 = 3R/2$.

Equation of state:

$$pV = RT$$

In process 1 – 2, $p = \alpha V$ is satisfied, which means $T \propto V^2$, therefore $T_2 = 4T_1$, $p_2 = 2p_1$, $\alpha V_1^2 = RT_1$. Let's write the relationship from the condition:

$$C_0 + aT_2 = C_0 + 4aT_1 = 1.5(C_0 + aT_1),$$

from which it follows:

$$C_0 = 5aT_1$$

Heat is supplied only in section 1 – 2 (the temperature monotonically increases), and is removed only in section 3 – 1 (temperature decreases monotonically). Let's find the supplied heat:

$$dQ = C_V dT + p dV = (C_0 + aT)dT + \alpha V dV$$

After integration we get:

$$Q_+ = C_0 \cdot (4T_1 - T_1) + a \cdot \frac{16T_1^2 - T_1^2}{2} + \alpha \cdot \frac{4V_1^2 - V_1^2}{2} = \frac{9}{2}C_0T_1 + \frac{3}{2}RT_1 = \frac{33}{4}RT_1$$

Now we find the work of the gas: in the section 1 – 2 $A_{12} = \frac{3}{2}RT_1$, section 2 – 3 is isochore, therefore $A_{23} = 0$, in the last section at a pressure p_1 the volume changes from $2V_1$ to V_1 , therefore $A_{31} = -p_1V_1 = -RT_1$. The total gas work is $A = \frac{RT_1}{2}$, the efficiency is found as

$$\eta = \frac{A}{Q_+} = \frac{2}{33}$$

Answer: \textbf{Answer:}

$$\eta = \frac{2}{33}$$

A6

0.50 pts

Let us consider an adiabatic process performed with a gas. Derive the relationship between infinitesimal increments of volume dV and temperature dT . Express your answer using C_0 , a , T , V , R .

Let us equate to zero the small increment of heat:

$$dQ = C_V dT + p dV = 0$$

From the equation of state:

$$p = \frac{RT}{V}$$

From here we have:

$$(C_0 + aT)dT + \frac{RT}{V}dV = 0$$

Answer: \textbf{Answer:}

$$(C_0 + aT)dT + \frac{RT}{V}dV = 0$$

A7

0.90 pts

From the relation in the previous paragraph, obtain an adiabatic equation connecting T and V of the form

$$f(T, V) = \text{const},$$

into the function f можым also входумь C_0 , a , R .

Dividing the expression from the previous paragraph by T , we get:

$$\left(\frac{C_0}{T} + a\right)dT + \frac{R}{V}dV = 0$$

Integrating, we get:

$$f(T, V) = C_0 \ln \frac{T}{T_0} + R \ln \frac{V}{V_0} + aT = \text{const}$$

Here T_0 , V_0 are arbitrary constants to ensure a dimensionless value under the logarithm. Omitting this formality, let's present the answer through the data in the quantity condition:

Answer: \textbf{Answer:}

$$f(T, V) = C_0 \ln T + R \ln V + aT = \text{const}$$

A8

0.40 pts

Consider the Carnot cycle 1234, consisting of isotherms 12 and 34 and adiabats 23 and 41. Express the volumes V_3 , V_4 in terms of V_1 , V_2 , temperatures on isotherms $T_+ = T_1 = T_2$, $T_- = T_3 = T_4$ and gas parameters C_0 , a , R .

Let's write the expression from the previous paragraph for the adiabatic 2 – 3:

$$C_0 \ln T_+ + R \ln V_2 + aT_+ = C_0 \ln T_- + R \ln V_3 + aT_-,$$

which transforms into:

$$R \ln \frac{V_3}{V_2} = C_0 \ln \frac{T_+}{T_-} + a(T_+ - T_-),$$

from which it follows:

$$V_3 = V_2 \left(\frac{T_+}{T_-} \right)^{\frac{C_0}{R}} \exp \left[\frac{a(T_+ - T_-)}{R} \right]$$

Similarly, we write for the adiabatic 4 – 1:

$$C_0 \ln T_- + R \ln V_4 + aT_- = C_0 \ln T_+ + R \ln V_1 + aT_+,$$

which transforms into:

$$V_4 = V_1 \left(\frac{T_+}{T_-} \right)^{\frac{C_0}{R}} \exp \left[\frac{a(T_+ - T_-)}{R} \right]$$

Answer: \textbf{Answer:}

$$V_3 = V_2 \left(\frac{T_+}{T_-} \right)^{\frac{C_0}{R}} \exp \left[\frac{a(T_+ - T_-)}{R} \right]$$

$$V_4 = V_1 \left(\frac{T_+}{T_-} \right)^{\frac{C_0}{R}} \exp \left[\frac{a(T_+ - T_-)}{R} \right]$$

A9

0.80 pts

By direct calculation (without using a general formula), show that the efficiency of the Carnot cycle is the same as for an ideal gas with constant heat capacity.

Heat can be supplied and removed only on isotherms (at 1 – 2 it is supplied, at 3 – 4 it is removed).

$$dQ = dU + dA = dA = p dV$$

($dU = C_V dT = 0$ since the process is isothermal)

$$pV = RT,$$

then

$$dQ_+ = RT_+ \frac{dV}{V},$$

$$Q_+ = RT_+ \ln \frac{V_2}{V_1},$$

similarly:

$$Q_- = RT_- \ln \frac{V_4}{V_3}$$

From the previous paragraph:

$$\frac{V_3}{V_4} = \frac{V_2}{V_1}$$

Let us express the efficiency:

$$\eta_1 = 1 + \frac{Q_-}{Q_+} = 1 - \frac{T_-}{T_+}$$

Efficiency of the Carnot cycle:

$$\eta_c = 1 - \frac{T_-}{T_+},$$

from which it follows that the expressions for the efficiency coincide.

Answer: \textbf{Answer:} The efficiency of the cycle coincides with the efficiency of the Carnot cycle.

B1**1.00 pts**

Find the amount of heat that needs to be supplied to the gas during isochoric heating Q_V . Express your answer in terms of T_1 , R .

On linear section $C_V = C_0 + aT$. We obtain a system of linear equations:

$$\begin{cases} C_0 + aT_1 = 3R/2 \\ C_0 + 2aT_1 = 5R/2 \end{cases}$$

From this system it follows:

$$aT_1 = R$$

$$C_0 = R/2$$

Let us find Q_V :

$$Q_V = \int_{T_1/2}^{3T_1} C_V dT = \frac{3R}{2} \left(T_1 - \frac{T_1}{2} \right) + \frac{5R}{2} (3T_1 - 2T_1) + \int_{T_1}^{2T_1} \left(\frac{R}{2} + \frac{RT}{T_1} \right) dT$$

$$Q_V = \frac{3RT_1}{4} + \frac{5RT_1}{2} + 2RT_1 = \frac{21}{4}RT_1$$

Answer: \textbf{Answer:}

$$Q_V = \frac{21}{4}RT_1$$

B2**0.50 pts**

Find the amount of heat that must be supplied to the gas during isobaric heating Q_P . Express your answer in terms of T_1 , R .

$$C_P = C_V + R,$$

therefore:

$$Q_P = Q_V + R\Delta T = \frac{21}{4}RT_1 + \frac{5}{2}RT_1 = \frac{31}{4}RT_1$$

Answer: \textbf{Answer:}

$$Q_P = \frac{31}{4}RT_1$$

B3**1.80 pts**

Let the initial volume of gas be V_0 . What will the final volume be equal to if the temperature was increased adiabatically? Find the work done in this process. Express your answers using T_1 , R , V_0 .

If in an adiabatic process with a heat capacity linearly dependent on temperature $C_V = C_0 + aT$, the temperature varies from T_{1x} to T_{2x} , and the volume from V_1 to V_2 , respectively, then the volumes at the beginning and at the end are related as follows:

$$V_2 = V_1 \left(\frac{T_{1x}}{T_{2x}} \right)^{\frac{C_0}{R}} \exp \left[\frac{a(T_{1x} - T_{2x})}{R} \right]$$

The process can be divided into three linear processes with the parameters:

$$C_0 = 3R/2, a = 0, T_{1x} = T_1/2, T_{2x} = T_1$$

$$C_0 = R/2, aT_1 = R, T_{1x} = T_1, T_{2x} = 2T_1$$

$$C_0 = 5R/2, a = 0, T_{1x} = 2T_1, T_{2x} = 3T_1$$

Then at the end of process 1:

$$V_1 = \frac{V_0}{2\sqrt{2}}$$

At the end of process 2:

$$V_2 = \frac{V_1}{e\sqrt{2}} = \frac{V_0}{4e}$$

At the end of process 3:

$$V_3 = V = \frac{4\sqrt{2}V_2}{9\sqrt{3}} = \frac{V_0\sqrt{2}}{9e\sqrt{3}}$$

Let's find the work done by the gas in an adiabatic process:

$$A = -\Delta U = -Q_V = -\frac{21}{4}RT_1$$

Answer: \textbf{Answer:}

$$V = \frac{V_0\sqrt{2}}{9e\sqrt{3}}$$

$$A = -\frac{21}{4}RT_1$$